A learned-correction value function with turn-level lookahead for 2-player Catan

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1. Problem and constraints

Our agent plays 2-player, 10-VP Catan without player trading on the organisers' catanatron build. It plays against F (catanatron's one-ply `ValueFunctionPlayer`), O (a trained bot) and other teams. Three facts shaped every design choice.

- **Compute is tiny.** The clock is 300 ms plus 12.5 ms per decision, capped at 1.5 s per game. We measured a `copy()+execute()` at about 25 µs and F's evaluation at 65 µs, so a decision can afford roughly 150 positions. MCTS and depth-3 alpha-beta run out of clock in every game (problem statement, Section 7).
- **Hidden information is small.** The opponent's resource hand is exactly derivable; only their unplayed development cards are hidden. Because of this we sample one world per decision (perfect-information sampling, Long et al. 2010) and spend compute on other things.
- **F's evaluation is a tiered linear sum.** Public VP is weighted 3×10^{14}, production 10^8, reachable production 10^4, building spots 10^3, and hand and dev-card terms 1 to 100. This works almost like a lexicographic order: a term only matters between moves that tie on every higher tier.

2. Method

2.1 Fast exact re-implementation of F's evaluation

We rewrote `base_fn` to compute each feature once (F's version computes production features twice) and only for our own seat. It is bit-for-bit identical to `base_fn` on 1,740 random positions and 3× faster (22 vs 65 µs), and that speed buys the search below. With lookahead off, our agent reproduces F's results exactly (193/400 wins on the same seeds as F vs F).

2.2 Learned win model as a correction term (the main gain)

We recorded positions after every action, from both seats, in 16,000 F-vs-F games (1.04M positions), each labelled with the eventual winner. On these we fit a logistic regression over 50 features. For each player the features cover public VP, production per resource, resource variety, road-reachable production, open building spots, hand composition and size, longest-road length, knights played, dev cards held, settlements, cities and ports, plus whose turn it is and our hidden VP cards. The model predicts the eventual winner for 74% of held-out positions (split by game): 62% in turns 0–20 and 80% from turn 40.

Used on its own as the evaluation, the model is poor (26% vs F with one-ply). It cannot judge within-turn trade-offs such as spending a hand versus holding it, which F's hand-set terms do encode. We therefore use a **hybrid**: $V(s) = V_F(s) + \alpha\cdot\text{logit}\, p_{\text{win}}(s)$ with $\alpha = 10^6$. At that scale the learned term sits between F's production tier and its lower tiers. It never overrides a VP or a large production change, but it decides the many moves F treats as near ties: roads, bank trades, dev-card purchases, robber placement and ending the turn. Larger α (10^8) lets the model overrule production and loses to F (47%).

2.3 Same-turn lookahead

F is myopic inside its own turn. A bank trade that enables a settlement looks bad on its own, because the trade alone lowers the hand terms. For the top-4 moves by one-ply value we roll the rest of our turn forward greedily (up to 4 more own moves) and score each move by the best value reached before ending the turn. This cannot be fooled by the opponent, because the rollout stops when the turn passes.

2.4 Opening simulation

For each initial-settlement decision we take the 6 best candidates by one-ply value and simulate the rest of the placement phase. Both seats place greedily by our evaluation from their own side. We keep the candidate whose finished placement scores best. The 300 ms starting bank pays for this.

2.5 Time manager and safety

The per-decision budget is min(cap, 0.25·(cpu_left − 200 ms)), with a 150 ms cap in the opening. Below 200 ms the agent drops to one-ply with F's cheap evaluation (from v5; v2–v3 used the hybrid here), and below 15 ms to a fixed move-type priority with no copying. Every decision is wrapped: on any exception it falls back to one-ply, then to index 0. No submission has had a fault. The only over-clock game (v3, 1 in 1,000) came from a very long game, and v5's cheaper reserve mode removed it.

2.6 In-run opponent adaptation (built, then dropped)

One agent process plays a whole match against a single opponent, and adapting is allowed. v4 therefore held several complete strategies ("arms"): the linear model, the MLP and a 64-unit MLP. Before each game it picks an arm by Thompson sampling from a Beta posterior over that arm's wins. It labels its own games: a game counts as won when our chosen move produced the 10th VP. Against F, where all arms are about equally strong, five processes of 200 games each scored 0.671 (v2: 0.684), so exploration costs nothing measurable. The self-labelling counts 61% wins against the true 67%, because it misses wins completed by forced moves. That bias is not guaranteed to be equal across arms, since arms may win in different ways. The tournament below shows v4 tied with v3 (0.500) and no better against any opponent, because arms that differ only in the predictor on top play too alike to be worth adapting between. We dropped the bandit for v5.

3. Experimental protocol

All local numbers come from the kit's own harness (`catan_kit.match`), run in parallel, with each board played from both seats with the same dice. Tuning used dev boards (seed 990000), and every claimed gain was re-checked on separate holdout boards (seed 700000). We required ≥1,000 games per arm (binomial 95% CI ±3 pp). Our search stops on wall-clock deadlines and the laptop's speed varies, so repeat runs of the *same* agent on the *same* boards differ by up to 2.5 pp (0.678 vs 0.654 for v3). The combined resolution is therefore about ±4 pp: we treat smaller differences as ties unless they replicate on both board sets. Weight tuning used its own boards, and model training used separate games. Every run is logged in `tools/results.csv`, 60+ runs in all.

4. Results

Agent (vs F)	Dev (1000)	Holdout (1000)	CPU/game
F vs F (sanity)	0.483 (400)		223 ms
F evaluation + same-turn lookahead	0.517		419 ms
Hybrid, one-ply	0.573 (400)		222 ms
Hybrid + lookahead (v1)	0.643 (400)	0.622	346 ms
Hybrid + lookahead, no opening simulation	0.642	0.640	361 ms
v2 = hybrid + lookahead + opening simulation	**0.684**	**0.672**	455 ms
v2 without the learned term (ablation)	0.517		419 ms
v2 without lookahead (ablation)	0.600		351 ms

Table 1. Win rate against F (95% CI ±0.03 at 1,000 games, ±0.05 at 400). CPU is on a laptop, where F itself uses 186–223 ms against 158 ms on the scoring machine. The learned term is worth 16.7 pp, lookahead 8.4 pp and the opening simulation 4.2 pp. The parts interact: lookahead alone adds only 3 pp to F, but 7 pp to the hybrid.

Scoring machine, public boards	vs F	vs O	Mean	CPU mean / worst	Over clock
v2 (linear win model)	0.649	0.487	0.568	368 / 1029 ms	0
v3 (MLP win model, otherwise identical)	0.698	0.505	0.602	450 / 1501 ms	1
mixed8000_extra *[TODO: describe]*	0.694	0.538	0.616	484 / 1453 ms	0
v5 (= v3 + cheap reserve mode)	0.680	0.480	0.580	456 / 1412 ms	0

Table 2. 1,000 games per cell, 95% CI ±0.03. Locally, v2 and v3 were equal against F (0.684/0.672 vs 0.678/0.667). On the public boards v3 is 5 pp better against F, which at our ±4 pp resolution is most likely board

luck, not a real gain. Both are roughly even with O. v3 and v5 differ only at the very end of long games, yet the portal puts them 1.8 pp apart against F and 2.5 pp apart against O: that is the portal's own noise. One upload is a single ±3 pp sample, and the local-to-public differences (−3.5 pp for v2, +2 pp for v3) are the same size as the differences between our own dev and holdout boards. The team machine (Linux, official script) gave v2 0.700 ± 0.064 over 200 games.

Where the CPU goes. Cutting lookahead from the top 4 moves to the top 2 saved only 8% of CPU (450→414 ms) and lost 2.6 pp, so the lookahead is not the cost. The cost is the one-ply pass over every legal move with the MLP hybrid (~100 µs per move) and, in the worst case, very long games. Past 96 decisions the clock stops growing, yet each decision still costs about 1 ms to receive the game state. v3's single over-clock game came from this. The fix is to fall back to F's cheap evaluation once the reserve is reached.

How v2 wins. Over 300 games vs F, v2 ends with longest road in 69% of games against 30% for F, and largest army in 28% against 3%. It buys 3.3 dev cards per game against 0.9, while building slightly *fewer* settlements and cities (4.0 vs 4.5). Longest road decides games: v2 holds it in 92% of its wins but only 31% of its losses. The learned term moved play towards the two 2-VP awards that F's tiered weights almost ignore. We never hand-coded this.

Robustness tournament. Other teams will mostly field F-like agents, so we built five F variants with re-tiered weights: build-heavy, dev/army-heavy, road-heavy, city-saving, and catanatron's tuned "contender". v2, v3 and v4 each played all of them, plus F, F with lookahead, v1, a learned-only agent, a mixed-data MLP, and each other. That's 36 matches of 500 games on identical boards (18,000 games). Scored as the group stage scores us, by the mean over the worst half of a 10-opponent F-like field:

	Overall	Worst half	Weakest matchup	Head-to-head
v2	0.683	0.624	mixed MLP 0.502	v2–v3 0.482
v3	**0.698**	**0.644**	v1 0.538	v3–v4 0.500
v4 (bandit)	0.685	0.623	mixed MLP 0.534	v2–v4 0.496

Table 3. 500 games per match (CI ±0.044). Against every F variant all three versions score 0.66–0.73 (0.89–0.92 vs the road-heavy variant), so no F-like style exploits us. The weakest matchups are our own learned variants.

5. What did not work, and why that matters

Attempt	vs F	Lesson
catanatron AlphaBeta, depth 2 (expectiminimax on F's evaluation), no clock	0.410	Deeper search does not fix a weak evaluation
Wider/deeper lookahead (top 8, depth 7); opening with 10–16 candidates	0.63–0.69	Search saturates quickly
Chance moves averaged over 4 samples	0.658–0.663	More CPU, no gain
Retraining on own games: mixed (DAgger-style); identity-controlled; pure v3 self-play	0.562 / 0.640 / 0.442†	Win labels from own play hurt, three times
Same self-play data, final VP-margin target instead of win/loss	0.515† / 0.668	Margin labels +7 pp over win labels on identical data, but only a tie with v3
F-vs-F data, VP-margin target	0.485† / 0.663	Tie
In-run bandit over 3 predictors (v4); exhaustive winning-line search at ≥7 actual VP	0.500† / 0.499†	Tie
MLP win model (val log-loss 0.472 vs 0.491 linear)	0.667–0.678	Better prediction did not mean better play
Hand-designed strategy terms: army race, zero-sum VP, road race, blocking, hidden-VP	0.50–0.69	Neutral or harmful

valuation		
Dedicated robber (exact steal expectation) and discard logic	0.663–0.679	Neutral
Opening model trained on 32k end-of-placement positions	0.644–0.693	No gain on holdout
CEM weight search, 200–240 games per candidate	flat	Inconclusive at ±7 pp noise per candidate

† head-to-head against v3 (1,000 games, same boards); other numbers are against F.

- **The evaluation, not search depth, is the bottleneck under a hard compute cap.** Expectiminimax with F's evaluation loses to one-ply F. A small correction term from outcome data gave our biggest gain.
- **The model steers play, but better prediction did not mean better choices.** We checked whether the learned term actually changes moves. Over 5,721 decisions in 120 games, our final move differs from F's in 47% of decisions (one-ply hybrid alone: 45%): roads 70%, bank trades 64%, discards 55%, ending the turn 46%, opening 32%, settlements 22%, robber 19%, cities 12%. The plateau is not because the term cannot act. Improving win *prediction* (log-loss 0.491→0.472) did not improve the *ranking of sibling moves*, which is what play needs. A model trained on outcomes ranks positions across games, not moves within a position.
- **Learned corrections should be scaled into the gaps of a hand-built order, not replace it.** With α too large the model overrules production and play gets worse. With α at about 10^6 it settles the moves the hand-built order leaves tied.
- **Labels matter more than data volume or features.** Win/loss labels from our own games hurt every time (mixed, identity-controlled, pure self-play: 0.44–0.56). In self-play the winner is decided by small edges, so a game outcome says little about any one position. On the same self-play data, regressing the final VP margin instead of the winner recovered 7 pp head-to-head. But no label, data source or architecture beat the original F-vs-F MLP by more than our ±4 pp resolution.
- **Hand-coded strategy hurt once a learned term was present.** The model already prices the army and road races; adding them again double-counts.
- **Noise discipline.** Several apparent +2–3 pp gains on 400–600 games vanished on 1,000 holdout games. We report only changes confirmed on both board sets.

References and public material used

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- Libraries: numpy, scikit-learn (training only). Code written with the help of an LLM coding assistant (Claude), as the rules permit. All numbers come from our own runs.